Case 2

Done	
Status	Cases

1. Computational psychiatry: problem and aim

Psychiatry diagnoses by symptoms, not mechanisms; computational psychiatry builds mathematical models of disease processes so that an individual patient's mechanism can be inferred from their data – a **computational assay**.

- **The problem**
  - Nosology (DSM/ICD) = syndromes defined by reported symptoms and signs → treated as categorical entities
  - But each label is a **spectrum disease**: same diagnosis, different underlying mechanisms
  - No test reveals the individual mechanism → **trial-and-error** pharmacotherapy
  - Knowing a drug's molecular action does not explain its effect on circuits and information processing
- **The aim: computational assays**
  - Models at two levels: **physiological** (circuits, synapses) and **computational** (information processing)
  - Applied to behaviour or neuroimaging of a single patient
  - Goal is NOT to reproduce DSM categories (that would copy labels lacking predictive validity)
  - Goal IS to detect the primary abnormality → split spectrum diseases into mechanistic subgroups → choose targeted treatment

Computational assay = a 'blood test' for mechanisms:

1. **Model** of disease mechanism is built and validated → defines what can be measured
2. **Data acquisition** from the patient (neuroimaging, electrophysiology, behaviour; behaviour could even be collected at home) → raw input
3. **Data analysis** with standardised software: model inversion, model selection, model averaging → estimate of the patient's mechanism
4. **Detection of physiological subgroups** across patients → patients grouped by inferred mechanism, not by symptoms
5. **Individual treatment prediction** → result supports (not determines) the clinical decision

- **Two model families** used for assays (both are generative models, §3)
  - Computational learning models of behaviour (e.g. HGF, §5) → what a neuromodulator *encodes* computationally
  - Biophysical circuit models of neuroimaging data (DCM, §7) → how a drug or disease changes *connections/synapses*
  - Best: combine both – knowing which computation needs which neuromodulator, and when, helps disentangle modulators that are otherwise hard to separate physiologically
- **Behavioural vs. neurophysiological assays**: app-based behavioural tasks are cheap and scalable but give less precise inference on neuromodulatory systems than neuroimaging
- **Ideal paradigm**: tasks sensitive to several neuromodulators at once (or a battery of tasks) → status of several systems per patient

2. Why neuromodulators: from physiology to computation

Most psychiatric drugs act on dopamine (DA), acetylcholine (ACh), serotonin (5-HT) or noradrenaline (NA); because these transmitters are broadcast widely and slowly, they are thought to encode **context variables** that set how circuits process information and learn.

- **Why they are the first-rank target for assays**
  - Major disease theories rest on them: 5-HT – depression; NA – ADHD; DA and ACh – schizophrenia
  - Most drugs alter their synthesis, metabolism or receptor binding
- **Modulatory vs. fast transmitters**
  - Fast (glutamate, GABA) → ionotropic receptors → directly change membrane potential
  - Modulatory → mostly **metabotropic** (G-protein-coupled) receptors, often **extrasynaptic** → do not directly change membrane potential

Neuromodulator action = changes the *rules* of a circuit rather than carrying the message:

1. Modulator binds metabotropic receptor → triggers G-protein intracellular cascade
2. Cascade changes **neuronal excitability** (e.g. via potassium currents) → gain of the neuron changes
3. Cascade changes **synaptic plasticity** on several timescales → ultra-short (phosphorylation of ionotropic receptors), short (receptor trafficking), long (gene transcription)
4. Result → how strongly and how flexibly a circuit responds and learns is altered

- **Anatomy → broadcast**
  - Source = a few subcortical/brainstem nuclei (DA: VTA/substantia nigra; ACh: basal forebrain + brainstem nuclei; 5-HT: raphe; NA: locus coeruleus)
  - Long-range projections + **volume transmission**: released from non-synaptic varicosities, diffuses to many receptors → one message reaches a wide target region, signals slow (tonic) changes
  - → plausible carriers of **global computational variables** (prediction errors, uncertainty/precision) that guide adaptive changes in processing mode and synapses
- **Tonic vs. phasic** activity patterns (DA, NA, 5-HT) → possibly different variables on different timescales (**multiplexing**)

3. Generative models and model inversion

A generative model is a probabilistic forward mapping from hidden causes (parameters, states) to observed data; inverting it means going backwards – inferring the hidden causes from the data.

- **Definition**: joint probability of data y and parameters θ = **prior** p(θ) × **likelihood** p(y|θ)
  - 'Generative' → you can sample θ from the prior, plug it into the likelihood and *simulate* synthetic data
- **Model inversion** = fitting the model to data → posterior over parameters (Bayes' rule)

$$p(\theta \mid y) \propto p(y \mid \theta)\, p(\theta)$$

- Inversion methods: sampling (MCMC) or approximation (variational Bayes); point estimates: MAP (max of posterior) or MLE (max of likelihood; = MAP with flat prior)
- Inversion also yields the **model evidence** p(y|m) → used to compare models (§8)
- **Two kinds of generative model in computational psychiatry**

	Computational learning model	Neural circuit model
Example	HGF, RW, Q-learning	DCM
Data explained	Behaviour (choices, RTs)	Neuroimaging (fMRI, M/EEG)
Hidden quantities	Beliefs, prediction errors, uncertainty, learning rates	Neuronal states, effective connectivity, its modulation
Neuromodulation enters as	Which computational variable a transmitter encodes	How a transmitter changes coupling / synaptic gain

- **Model-based fMRI** = the bridge between them
  1. Fit a learning model to behaviour → trial-wise trajectories of computational quantities (e.g. a precision-weighted PE)
  2. Enter these trajectories as parametric regressors in a GLM of fMRI → each trial's predicted 'amount' of the quantity

3. Voxel-wise test → regions whose BOLD tracks the quantity (e.g. midbrain ↔ PEs)
    - Limitation: BOLD in a nucleus reflects many cell types → cannot attribute a signal to one transmitter with certainty (motivates pharmacological designs, §10)

4. Learning from prediction errors

All learning models share one core: a belief is updated in proportion to the **prediction error (PE)** – outcome minus expectation – scaled by a learning rate; models differ in *what* is predicted and *how* the learning rate is set.

PE-driven update = how experience changes an expectation:

1. Agent holds expectation V → prediction for the next outcome
2. Outcome arrives → PE δ = outcome − V (positive: better than expected; negative: worse)
3. Learning rate α scales the PE → how much a single surprise counts
4. V ← V + αδ → expectation moves toward the outcome; repeated over trials, V converges on the average outcome

- **Rescorla–Wagner (RW)**: learns association strength V between cue and outcome λ

$$\delta = \lambda - V, \qquad \Delta V = \alpha\,\delta$$

- **Temporal difference (TD)**: predicts the *discounted sum of future rewards*; γ = discount factor (how much the future counts)
    - Recursive: value now = reward now + γ × value of the next state → PE can be computed at every time step

$$\delta_t = r_t + \gamma \hat{V}_{t+1} - \hat{V}_t$$

- **Q-learning**: same idea for *action values* Q(s,a); PE uses the **best** available next action (max Q)
- **SARSA**: PE uses the action **actually chosen** next

- **Softmax choice rule** = turns values into choice probabilities
  - Inverse temperature β: β → 0 random choice; large β → almost always the highest-value option
  - Low β = noise *or* exploration of low-value options

$$p(k) = \frac{\exp(\beta Q_k)}{\sum_i \exp(\beta Q_i)}$$

- **Key limitation**: a fixed α is suboptimal in a changing world → motivates precision-weighted, dynamic learning rates (§5)

Evidence: Dopamine reward PE (classic, defines the link between a computational variable and a neuromodulator). Single midbrain DA neurons were recorded in monkeys during cue → reward learning. DA neurons fire phasically to an *unexpected* reward (positive PE); after learning, the burst moves from the reward to the predictive cue and a fully predicted reward elicits no response; if a predicted reward is omitted, firing dips (negative PE) → DA neurons signal a signed TD prediction error.

5. Precision-weighting, uncertainty and the HGF

In a changing world a surprise should count more when one's current belief is uncertain; the Hierarchical Gaussian Filter (HGF) implements this by weighting every PE by a **precision ratio** (precision = inverse uncertainty), giving each individual a dynamic learning rate.

- **Three forms of uncertainty** (commuter example: predicting journey time)
  - **Irreducible** – randomness of outcomes, not reduced by learning (random station closure); highest at p = 0.5
  - **Estimation** – imperfect knowledge of the current probabilities; falls with learning (new route, gradually learned)
  - **Volatility** – belief about how fast the probabilities themselves change (Olympics: new crowds, rules change)
- **HGF levels** (as applied to a stimulus-transition task, §10)
  - Level 1 – x1: the stimulus transition that occurs (what happens next)

    - Level 2 – x2: the **transition contingencies** (probabilities, as a Gaussian random walk in logit space)
    - Level 3 – x3: **volatility** = the variance (speed) of the level-2 random walk
    - Each belief = mean μ + variance σ (uncertainty); precision π = 1/σ
- **Generic update rule** at every level i:

$$\Delta\mu_i \propto \psi_i \, \delta_{i-1}, \qquad \psi_i = \frac{\hat{\pi}_{i-1}}{\pi_i}$$

Precision-weighted PE = how much a surprise is allowed to change a belief:

1. Level below makes a prediction with precision π̂(i−1) → how reliable the incoming signal is expected to be
2. Input arrives → **PE δ(i−1)** = input − prediction
3. PE × **precision ratio ψ** = reliability of the input / precision of the current belief → **precision-weighted PE ε** (= the effective learning rate × PE)
    - Reliable input + uncertain belief → large update; noisy input + confident belief → small update
4. Belief μ(i) updated by ε → new estimate of the contingency
5. The change at level 2 itself produces a **contingency PE δ2** → passed up, weighted by its own precision ratio (ε3) → updates the **volatility belief μ3**
6. μ3 feeds back down: higher volatility → level-2 belief treated as less precise → larger learning rate for contingencies on the next trials
    - = how the model detects change points implicitly: after a context switch, PEs spike → volatility rises → learning rate rises → new contingencies learned fast; as the new context stabilises, μ3 and the learning rate fall again

- **Named PEs**: δ1 = **sensory PE** (from irreducible uncertainty; 'unexpected stimulus'), ε2 = precision-weighted sensory PE, δ2 = **contingency PE**, ε3 = precision-weighted contingency PE
- **Subject-specific parameters** (couple the levels; individual approximation to Bayes-optimality instead of an ideal observer)

- **ω** = constant (tonic) component of volatility → general speed of learning about contingencies = **tonic learning rate**
- **ϑ** = **metavolatility**, variance of the volatility random walk → how fast μ3 is updated; high ϑ = belief in an unstable world → **more variable (phasic) learning rate**

- **Perceptual model vs. response model** (two components of the HGF)
  - Perceptual model = how the agent learns: environment → beliefs (μ, PEs, uncertainty)
  - Response model = how beliefs are turned into observed behaviour; e.g. log reaction time as a linear function of trial-wise beliefs:

$$\log(RT)_t = \beta_0 + \beta_1\delta_{1,t} + \beta_2\varepsilon_{3,t} + \beta_3\mu_{3,t} + \beta_4\,\mathrm{PostError}_t + \zeta$$

- β0 = constant speed; β1–β3 = **sensitivity of responses** to sensory PE, contingency PE, volatility; β4 = post-error slowing; ζ = noise
  - → separates a drug's effect on **learning** (ω, ϑ) from its effect on **using beliefs to act** (β)
- **Precision and psychopathology**
  - **Schizophrenia – aberrant salience** formalised as a precision imbalance: abnormally high low-level precision and/or abnormally low high-level precision → chronically enhanced PEs to irrelevant events → delusions = hyper-precise high-level beliefs adopted to explain these PEs away
  - **Autism**: perception dominated by sensory detail = lack of regularising top-down abstractions, a putative precision deficit

6. Computational roles of the neuromodulators

Each modulator is proposed to encode a different learning variable – DA: prediction errors and precision; ACh: expected uncertainty within a context; NA: unexpected uncertainty when the context changes; 5-HT: temporal discounting and punishment – but functions overlap and the systems interact.

Modulator	Proposed computational variable	Effect of antagonist in the HGF study (§10)
Dopamine	Reward/sensory PE; precision of beliefs	No change in learning; responses less sensitive to volatility (↓β3)
Acetylcholine	Expected uncertainty; bottom-up vs. top-down balance	Slower contingency learning (↓ω), faster volatility updating (↑ϑ), responses less sensitive to all beliefs (↓β1–3)
Noradrenaline	Unexpected uncertainty; interrupt/reset signal	Faster volatility updating (↑ϑ) only
Serotonin	Discount factor; punishment; average reward/punishment rate	– (not tested)

Dopamine – prediction error and precision

- **Reward PE** (see Evidence in §4) → phasic DA = **teaching signal**: DA modulates NMDA-dependent plasticity → PE-driven synaptic change in task-relevant circuits → abnormal DA = maladaptive learning (addiction, depression, schizophrenia)
- DA neurons are heterogeneous → encode more than reward PE:
  - **Novelty** → motivational signal for exploration, or plasticity signal for memory encoding
  - **Uncertainty/precision**: slow changes in DA activity track uncertainty; midbrain activity tracks precision of beliefs even without reward → links DA to the value of information and the exploration–exploitation trade-off
  - **Sensory (low-level) PEs** in reward-free learning
- Precision view unifies 'DA for learning' vs. 'DA for motivation/vigor'
- In schizophrenia: hyper-confidence of acute psychosis ↔ DA-encoded precision; antipsychotics act via D2 blockade
- **HGF evidence**: DA antagonism did not change learning but reduced how strongly responses were adjusted to volatility estimates → DA **sensitises actions to beliefs about uncertainty** (cognitive switching after rule changes), possibly reading out ACh-dependent perceptual effects

Acetylcholine – expected uncertainty within a context

- Core idea: ACh **shifts weighting from top-down predictions to bottom-up input**
  - ↑ influence of thalamic input on cortex
  - ↑ gain of forward (ascending) relative to backward (descending) connections
- **Expected uncertainty** = known unreliability of predictive relationships within a familiar context → high ACh → top-down prediction trusted less → more weight on new sensory evidence
  - = the precision ratio of §5: precision-weighted PE about outcome probabilities
- **ACh – DA cooperation**: ACh depolarises DA neurons (lowers burst threshold); cholinergic brainstem input may be needed for DA neurons to compute PEs → together set the bottom-up/top-down balance in perceptual inference
- **HGF evidence**: ACh antagonism → **↓ω** (slower learning of contingencies: less weight on sensory evidence, more reliance on current beliefs) **and ↑ϑ**
  - Why ↑ϑ: if contingencies are learned poorly, chance fluctuations within a context look like violations → misattributed to **context switches** → volatility updated faster
  - = predicted by theory: ACh depletion → randomness of a context underestimated → chance events taken as context change (→ distractibility)
  - Converging: ↑ACh (AChE inhibitor) → ↑ω, i.e. faster belief updating
- **ACh balances** attribution of surprise to chance fluctuation *within* a context vs. a gross change *of* context
- **Psychosis links**
  - **Mismatch negativity (MMN)** = ERP to surprising stimuli in an oddball sequence; reduced in schizophrenia; depends on NMDA and ACh → ACh enhances short-term plasticity / postsynaptic gain of sensory cortex

- **Hallucinations**: too little cholinergic gain on bottom-up input → top-down predictions overweighted → percepts without stimulus

Evidence: Hierarchical PEs in midbrain and basal forebrain (decides ACh vs. DA roles). Participants learned changing cue–outcome probabilities during fMRI; HGF trajectories were used as regressors. The low-level precision-weighted sensory PE correlated with the dopaminergic midbrain, the higher-level precision-weighted PE about outcome probability (≈ expected uncertainty) with the cholinergic basal forebrain → two hierarchically related PEs map onto two different modulatory systems.

Noradrenaline – unexpected uncertainty and network reset

- **Unexpected uncertainty** = arises when the environment changes (context switch) → strongly unexpected observations → requires heightened sensory vigilance and discarding outdated top-down expectations
- **Interrupt / network-reset signal**: phasic LC release modifies ongoing cortical activity → enables new behavioural responses, accelerates learning
- **Phasic vs. tonic LC modes**
  - Phasic (low baseline, high bursts to relevant stimuli) → task engagement, filtering of distractors
  - High tonic → distractibility, attention shifting, exploration
  - Intermediate tonic level optimal (inverted U)
- Pupil diameter ≈ index of LC-NA activity → tracks surprise, perceived risk and stress
- **HGF evidence** (weak: did not survive correction for multiple comparisons, but confirmed by permutation tests): NA antagonism → ↑ϑ only, no change in ω or responses
  - Interpretation: NA **stabilises the volatility estimate** – keeps it from falling back to its low baseline so beliefs remain revisable in a volatile world
- Mapping to HGF: unexpected uncertainty (discrete context change) ↔ HGF's top level, the rate at which volatility changes (ϑ)

- Clinical: anxiety (high-trait-anxious individuals adapt worse to volatility, with reduced pupil responses), stress disorders, ADHD

Serotonin – discounting, punishment and opponency

- **Impulsivity** comes in forms that need different paradigms: **waiting** (preferring small-now over large-later), **stopping** (failing to inhibit an initiated response), **premature responding** (acting before enough information)
- Waiting impulsivity ↔ **temporal discounting**: ↑5-HT → higher discount factor γ → more patience; ↓5-HT → steeper discounting → impulsive choice (human findings; animal data conflict)
- **Opponency with DA** – proposed in contrasting forms:
  - Tonic 5-HT = average reward rate; phasic 5-HT = PE for punishments
  - Reversed: tonic 5-HT = average **punishment** rate → inhibits behaviour; tonic DA = average reward rate → invigorates behaviour
- Findings are diverse: 5-HT neurons are heterogeneous (respond to reward, punishment, specific stimuli, multiple timescales) → no unified theory; link to why SSRIs help depression still unclear

Evidence: Tryptophan depletion and the discount factor (shows how a model localises a drug effect to one parameter). Brain 5-HT was lowered or raised by diet; participants made delayed-reward choices fitted with a Q-learning model with discount factor, learning rate and inverse temperature. Only the discount factor changed: low 5-HT → steeper discounting → preference for small immediate rewards.

Overlap and interaction

- Same variable, several transmitters: DA and 5-HT both linked to impulsivity; DA and NA to exploration–exploitation
- Same transmitter, several computations (heterogeneous neurons, multiplexing)
- Systems interconnected (cholinergic interneurons regulate striatal DA release; 5-HT and glutamate inputs regulate DA neurons; co-release of several transmitters)

- → identifying one transmitter's contribution is an ill-posed problem → needs prior knowledge (a strength of generative models), biophysical models of the interactions, and pharmacological challenges in tasks sensitive to several transmitters

7. Dynamic causal modelling (DCM): models of neural circuits

DCM is a generative model of neuroimaging data: differential equations describe how hidden neuronal activity in a few regions is driven by experimental inputs and shaped by directed connections, and a forward model maps that activity onto the measured signal; inverting it estimates **effective connectivity** and how the experiment changes it.

- **Key parameters** (bilinear DCM for fMRI)
  - **Driving inputs** – experimental events that directly excite a region (e.g. visual stimuli → V1)
  - **Endogenous connections** – constant, context-independent coupling between regions
  - **Modulatory effects** – context-specific, additive changes in a coupling (e.g. threat strengthens AMG → V1)
  - Coupling = rate constant in Hz: strong connection → activity in source causes fast rise in target; inverse = time constant

DCM forward model = how an experiment generates the measured data:

1. **Driving input** enters a region → raises its neuronal state
2. **Endogenous connections** carry the activity to connected regions → spread through the network
3. **Modulatory input** (an experimental condition) changes the strength of specific connections → context-dependent network dynamics
4. Neuronal states → **hemodynamic forward model** (region-specific) → predicted BOLD

5. **Inversion**: parameters are adjusted until predicted and observed BOLD match → posterior estimates of each connection and modulation + model evidence

- **What 'causal' means**
  - Control-theory causality: how the state of one population causes the *rate of change* of another, and how external perturbations alter this
  - Not mere temporal precedence; inferred among **hidden neuronal states**, not among observations (contrast: Granger causality works on the measured signals, ignoring the nonlinear, region-variable hemodynamic mapping)
  - fMRI DCM has no conduction delays; timing differences are absorbed by region-specific hemodynamics
- **What modulatory effects mean physiologically**: fast changes in coupling = changes in **membrane excitability** and/or **short-term synaptic plasticity** (e.g. neuromodulator-induced phosphorylation of AMPA receptors) → DCM as a non-invasive window on neuromodulation
  - M/EEG DCMs are biophysically richer → can estimate e.g. dopamine-induced changes in AMPA/NMDA conductance, or cholinergic gain changes underlying the MMN
- **Requirements and design**
  - Needs **experimental perturbation**: without controlled input, standard DCMs predict a flat line → unsuited to resting state (except stochastic DCMs)
  - Ideal: factorial designs with a driving factor (stimulus) and a modulatory factor (task, attention, learning, threat) → DCM explains *how* an interaction arises
  - Regions chosen from a conventional GLM/SPM analysis + anatomy; time series = principal eigenvariate around subject-specific maxima
  - **Not circular**: GLM shows *that* a region responds; DCM compares *mechanisms* producing that response (e.g. a modulated afferent vs. changed self-inhibition vs. effect inherited from elsewhere)

    - Small networks (two or three nodes) often suffice: coupling can be polysynaptic, omitted sub-networks replaced by inputs
- **Linking DCM and learning models**: trial-wise PEs from a learning model used as modulatory inputs → tests which connections change with learning (plasticity driven by PEs)
- Anatomical priors (e.g. tractography) can inform connection priors; their benefit is tested by model comparison

8. Bayesian model selection and group inference

Models are compared by their **evidence** $p(y|m)$ – how probable the data are under a model, averaged over its parameters – which rewards accuracy and penalises complexity; parameters are only interpreted *within* the model (or models) that win.

- **Model evidence = accuracy – complexity**
    - Integrating over all parameter settings penalises extra parameters automatically (Occam's razor): a flexible model spreads its predictions thin
    - = a measure of **generalisability** (like cross-validation, but cheaper)
    - Approximations: free energy (DCM), BIC, AIC (§9)
    - Bayes factor = ratio of evidences; log-evidence difference of about 3 counts as strong evidence
- **Two types of inference**
    - On **model structure** (which architecture, which connections are modulated) → BMS is sufficient
    - On **parameters** (e.g. is this connection weakened by the drug?) → posteriors are conditional on the model → BMS first, then parameters
- **Group-level BMS**
    - **Fixed effects (FFX)**: same model assumed in every subject → group log-evidence = sum over subjects; suits basic physiological mechanisms
    - **Random effects (RFX)**: the best model may differ between subjects (patients, strategies) → estimates how often each model occurs in the population; robust to outliers

        - **Expected probability** = probability that the model generated the data of a randomly chosen subject
        - **Exceedance probability** = probability that this model is more likely than every other model tested
    - **Family comparison**: partition model space (e.g. all models with vs. without a connection) → compare families, pooling over the other features
- **Group-level parameter inference**
    - FFX: **Bayesian parameter averaging (BPA)** – combines subjects' posteriors into one group posterior
    - RFX: subject-wise MAP estimates → classical t-test/ANOVA (summary-statistic approach)
    - **Bayesian model averaging (BMA)**: parameter averaged over all models, weighted by posterior model probability → removes dependence on one winning model; useful when no model clearly wins or when comparing groups
- **What BMS can't do**
    - Only **relative**: says which of the *considered* models is best, conditional on the model space → define the model space carefully and compare all plausible models in one step
    - Evidence is defined for one data set → in fMRI-DCM, models with different regions cannot be compared (different data); in M/EEG they can
    - Absolute fit (R^2, 'variance explained') is not a validity test: it is itself a comparison against an implicit null model; poor fit makes models harder to distinguish but does not give false inference

9. Modelling behaviour in four stages

A model only tells you something about the mind if experiment and models can distinguish the hypotheses *in principle* (checked on simulated data) and the winning model reproduces the real behaviour; fitting and comparison alone never guarantee this.

- **Four uses of a behavioural model**

- **Simulation** – generate fake data → precise, falsifiable predictions
- **Parameter estimation** – summarise a person, compare individuals, groups, drugs
- **Model comparison** – which mechanism most likely produced the behaviour
- **Latent-variable inference** – hidden quantities (values, PEs) to relate to brain data

Stage 1 – Consider the right model (experiment, models, simulation)

- **Design the experiment first**: a model can only extract what the protocol makes visible
  - What scientific question / cognitive process?
  - Does the task actually engage that process?
  - Will its **signatures appear in simple statistics** of behaviour (e.g. vary working-memory load to see its effect)? No model-independent effect → modelling rarely helps
- **Design the models**
  - As simple as possible, but no simpler
  - **Interpretable** parameters (a negative learning rate signals a missing mechanism)
  - Include **competing hypotheses**, not strawmen; no favourite – let the data decide
  - Example set for a two-armed bandit: random responding (bias b) · noisy win-stay-lose-shift (ε) · Rescorla–Wagner + softmax (α, β) · choice kernel = tendency to repeat past choices (αc, βc) · RW + choice kernel
- **Simulate, simulate, simulate**
  - Define **model-independent measures** (e.g. p(stay) after reward vs. no reward; p(correct) early vs. late)

- Simulate across the **parameter range** → see how each parameter shapes behaviour (e.g. α and β *both* affect learning speed and asymptotic accuracy, and interact)
- Check that different models produce **qualitatively different** behaviour; if not, redesign the experiment

Stage 2 – Model check: is it interpretable? (fitting, parameter and model recovery)

- **Fitting by maximum likelihood**: find parameters that make the observed choices most probable

$$LL = \sum_{t=1}^{T} \log p(c_t \mid d_{1:t-1}, s_t, \theta, m)$$

- Practical pitfalls: start points with non-finite LL; rounding to 0 or ∞; parameters with no effect on the likelihood; **local optima** → run the optimiser from many random starting points and keep the best

Parameter recovery = can the fitting procedure find the true parameters in the best case?

1. **Simulate** data with known parameters → ground truth
2. **Fit** the model to the simulated data → recovered parameters
3. **Plot simulated vs. recovered** → should correlate tightly without bias; poor recovery = bug or underpowered task
4. **Check correlations among recovered parameters** → parameters that trade off are not separately identifiable (reparameterise, redesign, or report the trade-off)

- Use the parameter **range found in (or expected for) real data**; recovery often fails at extremes (e.g. very large β → near-identical behaviour)
- Can serve as a **power analysis** for group/drug differences
- Success = best case only; the real process is more complex than any model

Model recovery = can the comparison procedure tell the models apart?

1. **Simulate** data from every model (realistic parameter ranges) → known generator
2. **Fit every model** to every data set → model comparison score for each
3. Tabulate how often data from model A are best fit by model B → **confusion matrix** p(fit | simulated); ideal = identity
4. Invert with Bayes' rule → **inversion matrix** p(simulated | fit) → what a 'win' for a model actually lets you conclude about real data

- Recoverability depends strongly on parameters (more noise → models harder to distinguish)
- Scores penalise complexity, e.g. **BIC**; if the confusion matrix is poor, try another criterion (e.g. AIC) whose assumptions fit better

$$BIC = -2 \log \hat{L} + k_m \log T$$

- (k = number of parameters, T = number of trials; lower BIC = better)

Stage 3 – Fit real data (and validate)

- First **model-independent analysis** of the real data; no evidence of the targeted process → go back, don't model
- Fit and compare models; check fitted parameters lie in the range where recovery worked
- Model **'unimportant' parameters** (e.g. side bias, lapses) → variance otherwise attributed to noise is captured → better estimates of the parameters of interest
- **Validate (at least) the winning model**: model comparison is relative; validation tests whether the model captures behaviour in absolute terms
    - Simulate the model **with the fitted parameters** (posterior predictive check) → analyse simulated data exactly like the real data → reproduces key qualitative and quantitative effects?
    - Average likelihood per trial is of limited use (maximum unknown when behaviour is stochastic)

- Failure → back to the drawing board (better model, better task)

Evidence: Better fit ≠ captures behaviour (classic validation case). Data were generated by an RL agent in a reversal task and fitted with a choice-kernel model (repeats past choices) and a win-stay-lose-shift model. The choice kernel won on every fit measure, because RL choices are autocorrelated – yet, simulated with its fitted parameters, it ignores reward and cannot reverse. Only simulation exposes this.

Stage 4 – Interpretation and reporting

- **Only the winning, validated model** is analysed further (limits p-hacking)
- **Latent variables**: run the model with each subject's fitted parameters, **yoked to the subject's actual choices and outcomes** → trial-wise hidden quantities → regress onto pupil, EEG, fMRI (model-based fMRI, §3)
- **Parameters** as individual-difference measures (age, symptoms) or group/drug comparisons
- **Report**
  - Model selection: confusion matrix (model recovery), number of subjects best fit by each model, exceedance probabilities (group BMS), model-independent measures on simulated data
  - Parameters: full distributions, pairwise correlations, parameter recovery
  - Share raw data and code
  - Report all modelling steps even with an 'established' model – your task or population may differ
- **Loop back**: improve the model where it misses data; test its **predictions** in new tasks; advanced methods (MAP with priors, hierarchical Bayesian/MCMC, RFX model selection, joint fitting of RTs and neural data)

10. Application of the HGF: pharmacological fingerprints of uncertainty

By blocking NA, ACh or DA in healthy people and fitting one HGF to their reaction times, the study separated each transmitter's effect on **learning** (perceptual

parameters ω, ϑ) from its effect on **using beliefs to respond** (response β's).

- **Why this design**
  - Imaging activity in a nucleus can't be pinned to one transmitter (mixed cell types) → manipulate the transmitter directly
  - Three drugs + placebo in **one task and one model** → overlap between systems becomes visible (a single drug would make any effect look more specific than it is)
- **Task – probabilistic serial reaction time**
  - Four stimuli, each mapped to a button; respond as fast as possible
  - Which stimulus follows which is governed by a **transition matrix** that switches unsignalled between blocks
  - → all three uncertainties present: irreducible (probabilistic transitions), estimation (learning the current matrix), volatility (unsignalled switches)
  - Learning shows as **faster RTs to predictable** stimuli
- **Drugs (between subjects, double-blind)**: NA α1 antagonist (prazosin), ACh muscarinic M1 antagonist (biperiden), DA D2/D1 antagonist (haloperidol), placebo

Analysis pipeline = the four modelling stages applied (§9):

1. **Model-agnostic check** → RT falls as transition probability rises; within a block, predictable stimuli get faster and unpredictable ones slower; post-error slowing → participants learned the structure, and learning differed by drug (ACh group slowest)
2. **Perceptual model** (three-level HGF, §5) fitted per participant → ω (tonic contingency learning rate), ϑ (volatility learning rate)
3. **Response model** chosen by RFX model comparison among candidate regressors → log RT = constant + sensory PE δ1 + precision-weighted contingency PE ε3 + volatility μ3 + post-error
   - In placebo, all three belief quantities slowed responses → surprise, contingency updates and perceived instability all cost time

4. **Drug vs. placebo** comparisons for each parameter, controlled for alertness and body weight; validated by permutation tests, residuals and simulations with fitted parameters

- **Results → fingerprints** (interpretation in §6)
  - **NA−**: ↑ϑ only → NA stabilises volatility estimates; no effect on responses
  - **ACh−**: ↓ω and ↑ϑ, plus ↓ response sensitivity at all levels → ACh sets learning within contexts and thereby how surprises are attributed (chance vs. context change)
  - **DA−**: learning intact; ↓$\beta 3$ (responses less adjusted to volatility) → DA links action to uncertainty beliefs
  - DA− and ACh− also slowed responses in general (↑$\beta 0$)
- **Limits**
  - Drugs are receptor-selective, but systems interact → comparisons only against placebo, not between drugs
  - Receptor subtypes differ in function; tonic vs. phasic timescales not separated

11. Application of DCM: psilocybin and threat processing

Psilocybin (a 5-HT2A agonist) reduces amygdala responses to threatening pictures; DCM showed this is best explained by a weaker threat-induced **top-down influence of the amygdala on visual cortex**, not by stronger prefrontal control of the amygdala.

- **Question**: the GLM showed *that* the amygdala responds less under psilocybin; DCM asks *which connection change* produces this
- **Three candidate mechanisms**
  - **Top-down from PFC**: stronger inhibitory LPFC → AMG control (the mechanism found for the SSRI citalopram)
  - **Bottom-up from vision**: weaker V1 → AMG input

    - **Top-down to vision**: weaker AMG → V1 feedback, which normally tunes visual processing toward threat

- **Task**: amygdala reactivity task – match threatening or neutral pictures vs. shapes (baseline); placebo and psilocybin sessions in the same people

DCM analysis = the steps of §7–8 applied:

1. **Regions** from the GLM (threat/pictures vs. shapes): right V1, right amygdala, right lateral PFC → subject-specific time series
2. **Base model**: visual input drives V1; bidirectional endogenous connections V1 ↔ AMG ↔ LPFC → the shared architecture
3. **Model space**: threat modulates only bottom-up connections · only top-down connections · both (full model) → one model per hypothesis
4. **RFX Bayesian model selection** per drug condition → the **full model** wins under placebo and psilocybin → threat changes coupling in both directions; same mode of processing in both conditions
5. **Bayesian model averaging** → each subject's parameters weighted over all three models → drug effect can show up in any connection
6. **Repeated-measures ANOVA** (drug × connection) → only the **threat-induced modulation of AMG → V1 is reduced** by psilocybin; endogenous and input parameters unchanged; LPFC → AMG not increased

- **Interpretation**
    - Normally the amygdala, via feedback, tunes visual cortex toward threat → threat binds limited visual capacity → attention biased to negative stimuli (amplified in anxiety and depression)
    - Psilocybin weakens this feedback → visual cortex less threat-sensitive → capacity freed for positive information → shift of emotional bias toward positive
    - Likely a direct 5-HT2A effect in the amygdala, which then places less load on prefrontal regulation
    - Contrast with SSRIs: both normalise amygdala hyper-reactivity, but via **different network mechanisms** (SSRI: ↑ prefrontal down-regulation;

psilocybin: ↓ amygdala → visual feedback)

- **Limits**
  - Only three right-hemisphere regions; other prefrontal regulators (medial PFC, ACC, OFC, left LPFC) could be missed
  - Connectivity change did not correlate with mood or behaviour changes